

Mobile Phone Price Category Prediction

Task 4 Completion Report: Target Variable Identification

Project algorithm: Random Forest Classifier | Dataset: mobile_dataset.csv

1. Task Objective

Identify the dependent variable for the mobile phone price classification problem and define how the numeric source price is converted into categorical price classes.

2. Target Variable Decision

Item	Decision
Source column	price
Model target column	Price_Category
Target type	Categorical
Input features	Extracted phone specifications and connectivity features
Excluded input	price, because it is used to create Price_Category

3. Price-Category Rule

The project uses the same price-band rule established during dataset understanding. The rule creates a practical three-class classification target from the numeric price column.

Price condition	Price_Category
price < 15,000	Below 15,000
15,000 <= price < 30,000	15,000-29,999
price >= 30,000	30,000 and above

4. Verified Target Distribution

The target rule was executed against all 1,019 records in mobile_dataset.csv.

Price_Category	Records	Percentage
Below 15,000	288	28.3%
15,000-29,999	369	36.2%
30,000 and above	362	35.5%

5. Completion Status

Task TA-004 is complete. The dependent variable is documented as Price_Category, its derivation rule is defined, its class distribution is verified, and the original price column is marked for exclusion from model inputs to avoid target leakage.